

Human Review: Sharp Covering-Index Decay for Schatten Classes

Original automatic proof: `solution_packet.pdf`

Checked date: 19 June 2026

Status: Manually checked, focusing on Theorem 1.

Verdict: The proof of Theorem 1 appears correct.

Human Comments

I have checked Theorem 1 of the packet, which appears to be the new technical addition in the solution. The theorem proves the upper estimate

$$\Theta_{S_p(H)}(n) \leq n^{-1/p}$$

for the classical Schatten classes by constructing an explicit cover of the unit ball by the sets

$$A_k = \{T \in B_{S_p(H)} : \|P_k T P_k\|_p \leq n^{-1/p}\}.$$

The argument looks correct.

The covering step is clear. The finite pinching

$$E(T) = \sum_{k=1}^n P_k T P_k$$

is written as an average of unitary conjugates of T . Since Schatten norms are unitarily invariant and convex, $\|E(T)\|_p \leq \|T\|_p$. As $E(T)$ is block diagonal, this gives

$$\sum_{k=1}^n \|P_k T P_k\|_p^p \leq 1$$

for $T \in B_{S_p(H)}$, so at least one diagonal block has norm at most $n^{-1/p}$. Hence the sets A_k cover the unit ball.

The computation of the essential inradius of each A_k also appears correct. The easy inequality $\varrho(A_k) \geq n^{-1/p}$ follows from $n^{-1/p} B_{S_p(H)} \subseteq A_k$. For the reverse inequality, the proof uses a finite-codimensional subspace

$$Y = \bigcap_{j=1}^m \ker \Phi_j$$

and represents the functionals Φ_j by trace-duality operators B_j . One then chooses a rank-one operator $W = \theta_{u,v}$ supported in the infinite-dimensional tail of the k -th block, with

$$\langle B_j u, v \rangle = 0 \quad (j = 1, \text{dots}, m),$$

so that $W \in Y$. This is the key finite-codimensional tail argument.

The use of a finite-rank projection $R \leq P_k$ is sound. Since finite-rank compressions approximate $V = P_k T P_k$ in Schatten norm, one can choose R so that

$$(\|R V R\|_p^p + r^p)^{1/p} - \|V - R V R\|_p > n^{-1/p}$$

for every $r > n^{-1/p}$. Then RVR and the chosen rank-one perturbation rW live on orthogonal summands, so

$$\|RVR + rW\|_p^p = \|RVR\|_p^p + r^p.$$

The reverse triangle inequality then gives

$$\|P_k(T + rW)P_k\|_p > n^{-1/p},$$

and therefore $T + rW \notin A_k$. This proves $\varrho(A_k) \leq n^{-1/p}$, completing Theorem 1.

Minor Amendment

There is one small wording issue in the proof. The sentence saying that the left-hand side tends to

$$(\|V\|_p^p + r^p)^{1/p} > r > n^{-1/p}$$

should be slightly weakened. If $V = 0$, then

$$(\|V\|_p^p + r^p)^{1/p} = r,$$

not strictly larger than r . The correct statement is

$$(\|V\|_p^p + r^p)^{1/p} \geq r > n^{-1/p},$$

which is fully sufficient for the argument.

Relation to the Lower Bound

The matching lower bound in the later sharp-decay statement seems essentially to come from the work of Kania–Mařlany/Raja. Namely, the diagonal copy of ℓ_p inside $S_p(H)$ is contractively complemented, and the monotonicity of the covering index under contractively complemented subspaces transfers the known scalar lower estimate to the Schatten class. Thus the genuinely new technical point in the packet appears to be Theorem 1, i.e. the explicit Schatten upper-bound cover and the verification of the exact essential inradius of its pieces.

Review Outcome

Approve this packet as an apparently correct solution of the classical Schatten sharp-decay problem, with the minor inequality wording correction above. The proof of Theorem 1 appears correct, and the lower-bound side is consistent with the existing Kania–Mařlany/Raja machinery.